Modern Physics

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MODERN PHYSICS

1. SPECIAL RELATIVITY

$E = mc$
$E = (pc) + (mc)$

2. ATOMIC STRUCTURE
Rutherford, Bohr models
Quantum numbers:
n (principal)
l (angular)
m_l (magnetic)
m_s (spin)

3. NUCLEAR
Nucleus = protons + neutrons
Z = protons
A = Z + N

Mass defect

Decays:

Half-life:

4. FISSION
Heavy -> lighter + energy
U-235 + n -> Ba + Kr + 3n

5. FUSION
Light -> heavier
4H -> He + energy
Stars, hydrogen bomb